



SUPERIOR UNIVERSITY

Submitted By:

Name: Maryam Sharif
Roll number Su92-bsdsm-f23-038
Section: 4A
Date: 27, April 2025

Assignment: 11

Subject:

PAI (lab)

Submitted to:

Sir Rasikh Ali

Lab 11 Task: Differences between Key AI Concepts

1. LangChain

- **LangChain** is a framework that helps **connect LLMs (Large Language Models)** with external sources like databases, APIs, or documents.
 - It **builds advanced applications** (like chatbots or agents) by chaining together LLMs with tools, memory, and user inputs.
 - **In simple words:** It's a toolkit that makes LLMs smarter by giving them memory, knowledge, and decision-making skills.
-

2. RAG (Retrieval-Augmented Generation)

- **RAG** is a method where **LLMs** are combined with **retrieval systems** (like search engines or VectorDBs).
 - Instead of answering only from memory, RAG first **retrieves information** from external documents, then **generates** a complete answer.
 - **In simple words:** It first **fetches** the right info, then **creates** the response improving accuracy and freshness.
-

3. LLMs (Large Language Models)

- **LLMs** are AI models trained on **huge amounts of text data** to understand and generate human-like language.
 - Examples: GPT-4, Gemini, Claude, Qwen, etc.
 - **In simple words:** LLMs are **super smart text machines** that can chat, write, summarize, or translate.
-

4. FAISS

- **FAISS** (Facebook AI Similarity Search) is a **library** developed by Facebook AI to **search through vectors** very quickly.
- It helps find **similar documents or embeddings** efficiently.

- **In simple words:** FAISS is like a **super-fast search engine for AI data** stored as vectors.
-

5. Vector

- A **vector** in AI is just a **list of numbers** that represents complex things (like text, images, etc.) in a machine-understandable form.
 - These vectors capture the **meaning or features** of data.
 - **In simple words:** A vector is a **numerical version** of real-world data.
-

6. VectorDB (Vector Database)

- **VectorDB** is a **special database** that stores and manages **vectors** instead of normal rows and columns.
 - It helps with **semantic search** finding things that are similar in meaning, not just by keyword.
 - Examples: **Pinecone, Chroma, FAISS, Milvus.**
 - **In simple words:** VectorDB is a **memory storage** for smart searches based on meaning, not just words.
-

7. Generative AI

- **Generative AI** refers to AI systems that can **create new content** like text, images, music, videos, code, etc.
 - Examples: ChatGPT (text), DALL-E (images), MusicLM (music).
 - **In simple words:** Generative AI is **creative AI** that makes something new.
-

8. GANs (Generative Adversarial Networks)

- **GANs** are a type of **Generative AI** that uses **two neural networks** (Generator and Discriminator) that compete against each other to create **realistic new data**.
- Used for: fake images, deepfakes, art generation, etc.

- **In simple words:** GANs are like a **game between two AIs** one makes fake stuff, and the other tries to catch the fake both get better over time.
-